

ADCS Team — 3-Day MATLAB/Simulink Sprint

Sprint: Sunday, September 13 – Tuesday, September 15, 2026 | Live Demo: Tuesday, September 15, 2026

Goal: a demo-ready, closed-loop MATLAB/Simulink simulation — orbit + environment + 6-DOF dynamics + sensors + detumbling control + automatic handoff to pointing control + live 3D visualization — shown on the 15th. This supersedes the earlier Week 1 ADCS plan now that MATLAB access is confirmed.

Day-by-Day Tasks

Day	Date	Focus & Tasks
Day 1	Sun, Sep 13	Install the CubeSat Simulation Library (MATLAB Add-On Explorer → search "CubeSat") on top of Aerospace Blockset. Open the CubeSat Simulation Project template from the Simulink Start Page and run the default model as-is to confirm orbit propagation and the Simulink 3D Animation visualization work. Set your target orbit (e.g. ~500 km circular LEO) via the "Edit Initial Orbit and Attitude" block. Study how the Spacecraft Dynamics block and Attitude Profile block are wired — this is the scaffolding you'll extend, not rebuild.
Day 2	Mon, Sep 14	This is the core ADCS day — build the parts that are actually yours: (1) a disturbance-torque subsystem (gravity-gradient torque at minimum) feeding the Spacecraft Dynamics block; (2) a sensor subsystem — magnetometer, sun sensor, and gyroscope, each with realistic noise; (3) a custom B-dot detumbling control law driving magnetorquers, plus a mode-switch state machine that hands off to the library's built-in Earth-Pointing or Sun-Tracking mode once the tumble rate drops below a threshold. Run the full chain: aggressive random tumble → detumble → automatic handoff → pointing acquisition.
Day 3	Tue, Sep 15	Polish and prepare the demo. Wire up result logging and plots: angular velocity during detumble, pointing error convergence, magnetic field and B-dot torque profile, actuator effort. Confirm the Simulink 3D Animation window is demo-ready (orbit + attitude + pointing target visible). Run 2–3 scenarios (different initial tumble rates, with and without disturbance torques) and save the results. Write a short results summary and rehearse the live walkthrough before presenting.

Deliverables & Live Demo — Due Tuesday, September 15

- A single extended Simulink model (built on the Aerospace Blockset CubeSat Simulation Library) that includes: orbit propagation, environment models (Earth magnetic field + sun vector, via built-in blocks), full 6-DOF rigid-body dynamics (Spacecraft Dynamics block), a custom 3-sensor suite (magnetometer, sun sensor, gyroscope) with noise, a custom B-dot detumbling control law with magnetorquers, and an automatic mode-switch into the library's Earth-Pointing or Sun-Tracking mode.
- A working 3D animated visualization (Simulink 3D Animation) showing the orbit and attitude motion live.
- A result plot set: angular velocity during detumble, pointing error convergence after handoff, magnetic field / control torque profile, and actuator effort.
- A live demo on the 15th showing at least 2 different initial tumble-rate scenarios, including one with disturbance torques enabled.
- A short (1–2 page) results summary covering model architecture, assumptions, and what the plots show.
- Model and scripts committed to the shared git repository with a README explaining how to run it.

Notes

- Feasibility lever: don't build orbit propagation, environment models, or visualization from scratch — the CubeSat Simulation Library already provides all of it. Spend your 3 days on the parts that are genuinely your ADCS design: sensors, the detumble law, and the control mode-switch.
- Parallelize on Day 1: one sub-team installs/explores the template and sets up the orbit and disturbance-torque subsystem, another starts drafting the sensor noise models, and a third studies the Attitude Profile block and default pointing modes — converge into one integrated model by end of day.
- Keep a fallback scope in mind: a rock-solid detumble-to-pointing handoff with clean visualization is a strong demo on its own. If the custom control isn't fully stable by Day 3 morning, present that core loop rather than a shakier, more feature-packed version.
- Use Simulink Scope/Dashboard blocks for live plotting during the demo itself rather than only static screenshots — it reads as more polished and lets you show a second scenario on the spot if asked.